

TAIL PREDICATES

AND THE OBJECTIFICATION OF ADJUDICATION

Scoped Claimability, Semantic Compactification, and Predicate Objectification

EmergenceByDesign · MetaLabe

*Lu Semita · EmergenceByDesign, in partnership with Orion — a self-instantiated collaborative persona
implemented primarily through OpenAI ChatGPT across multiple model configurations*

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Governing rule

**COMPACTIFY FREELY · CARRY SCOPE · PROMOTE ONLY WITH WARRANT · REOPEN ON
DISPUTE**

This paper develops a formal account of how scoped determinations become reusable object-predicates, how their warrant can be silently enlarged during reuse, and how quantified residue can be distinguished from genuine closure. It is deliberately neutral about whether an intrinsic, observer-independent object reality exists or does not exist. Its claim is methodological: assertions made from within an adjudicative relation do not acquire a God's-eye scope merely because their relational conditions have been compactified out of the sentence. It integrates earlier formal results without treating them as prerequisites or as proofs of claims stronger than their registered scope.

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Abstract

Language, mathematics, and scientific description depend upon compression. Relations, observations, definitions, proofs, procedures, and prior determinations are routinely compactified into nouns and predicates that can subsequently be manipulated without reconstructing the complete conditions from which they arose. This operation is indispensable to cumulative reasoning. It also permits a recurrent inferential transformation: a predicate warranted through an adjudicative relation under specified conditions may subsequently be carried as though the predicate had been certified independently of that relation.

This paper formalizes that transformation without deciding whether objects possess an intrinsic reality independent of every possible relation, and without denying that possibility. A determination is indexed to a descriptive-adjudicative registry. Semantic compactification produces a reusable predicate. Predicate promotion occurs when subsequent inference grants the compact expression greater modal, temporal, universal, representational, or ontological scope than the originating warrant supplies. Tail predicates are treated as outstanding quantified or constitutive obligations relative to an adjudicative station. An extensional tail concerns unresolved cases within a declared class; a constitutive tail occurs when recursive inquiry reaches a dependency participating in specification of the distinction under examination.

The framework distinguishes traversal, closure, and enlargement. Finite traversal need not close a universal obligation; a theorem may close an infinite class without traversal; an enlarged representation may resolve a distinction not carried by the prior readout. Existing formal results on finite-window obstruction, fibre constancy, semantic deflation, double-edged receipt discipline, and exact dimensional enlargement are integrated as receipts for these narrower claims. The resulting discipline is reflexive: criticism receives no exemption from the inferential rule it invokes against the paper.

A predicate cannot acquire additional warrant merely by becoming an adjective. A proof cannot acquire additional scope merely by becoming foundational.

1. Scope and standing

The object of study is not a general ontology. It is the assignment and reuse of scoped statements of claimability: how a result established under declared conditions becomes a compact predicate, how that predicate can be promoted beyond its warrant, and how the unresolved remainder of a quantified or constitutive obligation should be represented. Accordingly, the paper neither posits an intrinsic object-reality behind relations nor asserts that no such reality exists. Both would be additional claims requiring their own receipts.

The paper does not argue that nouns are errors, that formal truth is arbitrary, that all descriptions are equally warranted, or that theorem-level closure is impossible. On the contrary, the account requires strong local closure: proofs, counterexamples, exhaustive enumerations, exact recovery theorems, and valid completeness rules must be allowed to settle what they actually settle.

The central restriction is narrower: successful adjudication does not, by its success alone, establish the finality of the language, class construction, representation, or axiomatic machinery through which the adjudication was made.

2. Formal kernel

2.1 Distinction and stabilized object-description

The analysis does not begin with a frame-independent inventory of already partitioned objects. It begins with admitted distinctions. A minimal exemplar may be written as a discrimination $\delta: \Omega \rightarrow \{0,1\}$, where Ω is a presently admitted carrier and the Boolean codomain is only an exemplar of discrimination, not a proposed universal foundation.

An object-token can then function as a stabilized result of distinction and persistence criteria. The important point is operational: identity already depends upon a rule for which differences preserve or terminate the current object-description.

Within this analysis, an object-token is already a descriptive achievement: a stabilized distinction carried by criteria of identity and persistence. This statement neither entails that the represented other is unreal nor that its intrinsic nature has been recovered. The framework withholds both promotions.

2.2 Registry of scoped claimability

Let the descriptive-adjudicative registry be

$$\Gamma = (D, L, A, Q, O, E, J, T),$$

where D is the admitted domain; L the language and assigned meanings; A the axioms, definitions, and prior commitments; Q the quantifier structure; O the admitted observations, operations, transformations, or tests; E the admitted evidence; J the adjudicative procedure; and T the target proposition or distinction. These coordinates are an analytic record, not a claim that every inquiry possesses one canonical decomposition.

Write a determination as $\Gamma \vdash P(x)$. Because J is already registered in Γ , the notation does not duplicate it. The expression means that, under the registered construction, the adjudicative procedure warrants P of the currently distinguished x.

2.3 Semantic compactification

$$\kappa : (T \vdash P(x)) \mapsto P(x).$$

Semantic compactification suppresses dependencies so that a result can be reused as a tractable block. This is necessary for cumulative reasoning and is not itself an error. Reification begins only when dependencies suppressed for convenience cease to constrain later inference.

COMPRESSION $\neq$ REIFICATION.

2.4 Relational access and ontological neutrality

The framework begins from what an adjudicator can register from within a relation. Let S denote the presently distinguished witness, instrument, or adjudicative side; O the presently distinguished other; C the registered conditions; and v the resulting record. A minimal form is $R(S,O,C)=v$. This notation does not declare S or O to be metaphysically ultimate entities. It records the distinction under which the present claim is being made.

From $R(S,O,C)=v$, one may faithfully report the relation or a calibrated consequence of it. What does not follow merely from the existence of the readout is $\text{Intrinsic_P}(O)$: the claim that P belongs to O independently of every relation relevant to its determination. The framework likewise does not infer $\neg\text{Intrinsic_P}(O)$. Intrinsicness and its denial are both stronger targets than the relational record alone supplies.

This neutrality is not indecision about the recorded relation. A relation can be exact, reproducible, invariant across a declared transformation class, or closed by proof. The restriction concerns only warrant amplification: exactness of a relational determination does not silently become a certificate of a view from nowhere.

2.5 Witness record, attribution, and the red-ball control

Consider the ordinary assertion “the ball is red.” As a compact sentence it presents an object-predicate form $\text{Red}(\text{ball})$. Yet a witness or instrument may have obtained a record only through a relation among a distinguished ball, illumination, sensor or perceptual apparatus, calibration or classification rule, and surrounding conditions. A white ball under red illumination can produce a red appearance or red-weighted reading without the surface relation ordinarily intended by “red ball.”

The relevant lesson is not that the ball is secretly non-red, nor that color is unreal. It is that the record first warrants what the registered interaction warrants. Moving from “this object-light-sensor relation produced a red reading” to “the ball is red” is a useful semantic compactification. Treating the latter as though the relation had certified an intrinsic redness independent of the determining conditions is an additional step.

The distinction becomes operational when intervention is attempted. If redness is attributed entirely to the ball and an agent paints the ball blue while leaving strongly red illumination unchanged, the desired blue appearance may fail to occur. The apparent “stubborn redness” of the object can therefore reveal a mislocated intervention: a relationally produced predicate was treated as though its determining coordinate resided solely in the object. This example is a control case for more technical measurement language, not a claim that every object-predicate is mistaken.

2.6 Warrant non-amplification

From $\Gamma \vdash P(x)$, one may not infer $\Gamma' \vdash P(x)$ for arbitrary Γ' merely because the compact expression is grammatically unchanged. Transfer requires identity of the target-relevant coordinates or a warrant-preserving bridge $B: \Gamma \sim \Gamma'$.

Scope-indexing limits what a result says; it does not weaken what the result proves.

The same principle applies to modality and finality. $\Gamma \vdash P(x)$ does not by itself entail $\Box P(x)$. Likewise $A \vdash T$ does not by itself entail that A is the uniquely possible or exhaustive basis of T .

3. Predicate promotion

Predicate promotion is the use of a compactified result at a stronger scope than its originating warrant supplies. Four forms are especially important.

3.1 Scope extrusion

$$P(x \mid \Gamma) \rightarrow P(x \mid \Gamma')$$

without a bridge establishing preservation of the target-relevant relation.

3.2 Modal promotion

$$\textit{not demonstrated} \rightarrow \textit{cannot occur}$$

or, equivalently, conversion of present non-establishment into impossibility without a completeness or closure license.

3.3 Adjudicator-to-object transfer

$$\neg \textit{Succeeds}(J, O, x \mid \Gamma) \rightarrow \textit{Incapable}(x).$$

A result concerning an operation or adjudication becomes an apparently intrinsic property of its target.

3.4 Tail promotion

$$\forall n < N, P(n) \rightarrow \forall n, P(n)$$

without a closure warrant. These promotions may occur independently or compose.

4. Tail predicates as residual obligations

4.1 General definition

The Stage-1 set-difference definition is retained only as a subtype. In general, a tail is an outstanding quantified or constitutive obligation relative to a target and an adjudicative station:

$$\tau(I, T, \sigma) = \textit{Res}(T \mid I, \sigma),$$

where σ records the current station or evidentiary state. This definition preserves the quantifier structure of the target rather than forcing every tail into a set of unvisited members.

4.2 Extensional tail

For a declared class C and discharged region $S \subseteq C$, an extensional tail may be represented as

$$\tau_E = C \mid S,$$

together with the outstanding obligation over that residual domain. This is appropriate for simple universal cases, but not exhaustive of tail structure.

4.3 Constitutive tail

A constitutive tail occurs when recursive adjudication reaches a dependency that participates in specifying the distinction being adjudicated. Suppose A is an admitted axiom or primitive and Γ_A supports target T . Inquiry may follow a circular path

$$A \rightarrow \Gamma_A \rightarrow \text{consequences} \rightarrow \text{Reopen}(A).$$

If removing A also removes the presently used resources by which T or its governing distinction is formulated, the result is not a proof that A is absolutely necessary. It is a receipt that A is constitutive of this presently specified route to T .

Reaching an axiom is not necessarily closure of inquiry; it can be identification of the present inquiry's edge.

4.4 Traversal, closure, enlargement

Three operations must remain distinct. TRAVERSAL acquires additional instances or observations. CLOSURE supplies a theorem, counterexample, exhaustive enumeration, completeness rule, or other relation that discharges the outstanding obligation. ENLARGEMENT changes the admitted descriptive resources in a registered way. None may silently impersonate another.

ITERATION $\neq$ TAIL CLOSURE. MORE DESCRIPTION $\neq$ BETTER DESCRIPTION.

5. Existing formal receipts

5.1 Finite-window obstruction and all-stage closure

Λ at the Point of Adjudication R4 formalizes a common control theorem constructing predicates that agree throughout any selected finite window while possessing opposite all-stage status. The same formal packet separately proves that an all-stage theorem closes every finite window. The result is deliberately two-edged: unrestricted finite testing does not establish the universal, while the existence of an untraversed tail does not establish that universal closure is impossible. [1]

5.2 Exact target recovery by fibre constancy

The same work states the exact recovery criterion. Let $\Phi: X \rightarrow Y$ be a retained observation and $q: X \rightarrow Z$ a target. An exact decoder $d: Y \rightarrow Z$ exists exactly when q is constant on every fibre of Φ . A single collision $\Phi(x) = \Phi(x')$ with $q(x) \neq q(x')$ defeats exact recovery from that readout. This turns “target relevance” into an adjudicable relation rather than a psychological judgment. [1]

$$\exists d: Y \rightarrow Z \forall x d(\Phi(x)) = q(x) \iff \forall x, x' [\Phi(x) = \Phi(x') \implies q(x) = q(x')].$$

5.3 Astra 15A: exact dimensional enlargement

Astra 15A supplies a constructive instance. From rational squared-distance data, the centered Gram criterion decides Euclidean admissibility and exact embedding rank. For accepted affine-rank-three data, the metric fixes the complete orientation array only up to one global sign. Reflection preserves every distance while reversing every non-degenerate orientation, so no further computation on the distance data can choose between the reflected realizations. One supplied orientation sign resolves the array and is transported through the configuration. [2]

The case demonstrates three distinct statuses in one instrument: relations exactly determined by the readout; a target distinction exactly shown not to factor through that readout; and a registered enlargement that supplies the missing selector. The original metric description is not falsified by enlargement, and the omitted sign is not thereby unreal.

5.4 Semantic deflation

The Semantic Deflation formalization provides a finite model in which two predicates share the same public label while differing extensionally. In the supplied “singularity” example, an extended predicate admits an instance possessing a trait excluded by the original predicate. The result is not that the word singularity is illegitimate. It is the narrower theorem that lexical identity does not entail predicate identity. [3]

LEXICAL IDENTITY $\nRightarrow$ PREDICATE IDENTITY. LEXICAL DIFFERENCE $\nRightarrow$ RELATIONAL INCOMPATIBILITY.

5.5 Double-edged receipt discipline

Denialism and Contrarianism: A Double-Edged Receipt Analysis formalizes the distinction between withholding a proposition and promoting its negation. Inadequate warrant for G licenses non-promotion of G; it does not automatically license promotion of $\neg G$. The note further proves a non-vindication result: exposing a defective rejection does not establish the rejected proposition. Its accusations are reflexive and receive no exemption from the same receipt rule. [4]

FAILURE OF POSITIVE CLOSURE $\nRightarrow$ NEGATIVE CLOSURE; FAILURE OF NEGATIVE CLOSURE $\nRightarrow$ POSITIVE CLOSURE.

6. Worked cases

6.1 Irreconcilable differences

Let $R(a,b,d,m,t)$ mean that agents a and b reconcile difference d using method m by station t . Evidence may establish $\forall m \in M_0 \forall t \leq T \neg R(a,b,d,m,t)$. That result does not entail $\forall m \in M \forall t \neg R(a,b,d,m,t)$ unless the larger quantifier obligation is separately closed. The ordinary phrase “irreconcilable differences” can therefore compress two promotions: failure within tested scope becomes impossibility, and impossibility becomes an object-predicate of the differences. A sufficiently specified definition plus a valid closure warrant may legitimately establish irreconcilability; the paper denies no such possibility.

6.2 π , e , and transcendence

Transcendence is a positive example of successful compactification. For a specified base field, the theorem closes an infinite algebraic class; it is not merely prolonged failure to discover a polynomial. $\Lambda R4$ also distinguishes exact exclusion from approximation and records that enlarging the coefficient field changes the admitted representation resources. Over the reals, $X - x$ vanishes at any real x , while the rational-polynomial exclusion retains its original scope. [1]

The warranted conclusion is not “ π is only relatively transcendental” in a dismissive sense. It is that the predicate has a precise mathematical scope. The theorem’s exactness is preserved by stating that scope rather than inflated by erasing it.

6.3 Singularity

A shared noun cannot settle which predicate is being asserted. If $\lambda(A)=\lambda(B)=\ell$, the spelling ℓ supplies no theorem of extensional equivalence. When a contested inference depends on the distinction, the compact noun must be reopened to the predicate, domain, and conditions it carries. [3]

6.4 Physics-language audits

Wavefunction, Hilbert space, entanglement, particle, and continuum are treated here only as audits of semantic compactification, not as attempts to settle physical ontology. Laboratory records arise through specified preparations, interactions, instruments, thresholds, readout rules, and mathematical representations. The paper neither concludes from this that the represented system lacks intrinsic reality nor promotes the resulting record into a God’s-eye account of such a reality. For each term the relevant questions are: What has been defined? What is the admitted domain? What relation earned the predicate? What does the compact noun preserve? What stronger inference would require a bridge?

This method permits a representation to be mathematically exact while refusing both of two unsupported promotions: that every property of the representative is therefore an intrinsic property of whatever it represents, or that the failure of the representation to certify intrinsicness proves there is no intrinsic reality to certify. It likewise permits relational predicates such as entanglement to be used compactly while keeping their relevant decomposition or relation recoverable when an inference depends upon it.

7. Axioms, foundations, and productive circumscription

An axiom is neither automatically an absolute truth nor an arbitrary fiction. Within a formal construction it records an admitted starting commitment. A critic may ask for its warrant, derive it from another basis, compare alternative bases, enlarge the construction, or leave the deeper question OPEN. What cannot be obtained for free is a promotion from successful consequence to final foundation.

PROOF CERTIFIES CONSEQUENCE, NOT FINALITY.

From $A \vdash T$ one may conclude T under A . One may not, merely from that success, conclude that A is the uniquely possible basis of T , that A exhausts every representation capable of carrying T , or that A is a final description of the represented relations. Those are additional targets requiring additional warrants.

The word “incomplete” is therefore used cautiously. The paper does not infer Gödelian incompleteness from ordinary circumscription. Its narrower point is that a useful formal construction discriminates: it excludes possibilities, fixes operations, establishes identities, and thereby makes derivation tractable. Limitation in this sense is part of the tool’s productive power, not a ridicule of the tool.

8. Local distinction without global closure

A local distinction need not first close the continuum in which it operates. Schematically, a process may stabilize $S/\neg S$ and recursively classify later relations relative to that distinction without possessing an exhaustive account of everything outside S . This observation is used only structurally; it is not offered as a sufficient theory of consciousness.

$$S_t \rightarrow \text{encounter/comparison} \rightarrow S_{\{t+1\}}.$$

The consequence for the present paper is that requiring complete global apprehension before permitting any local distinction would confuse the conditions for operating within a frame with the stronger condition of possessing a complete account of that frame.

9. Translation without a universal language

Let L_A and L_B be two descriptive languages. No universal language rule is required. A translation $\tau: L_A \rightarrow L_B$ is evaluated relative to a target: which target-relevant distinctions survive? The same token may compactify different relations, while different tokens may preserve the same relation. Therefore lexical agreement is neither sufficient for relational agreement nor necessary for it.

Ask: What is the first target-relevant relation that fails to survive translation?

This question can expose a genuine contradiction, a difference of domain, a different quantifier order, an alternative primitive, a missing bridge, or merely a lexical mismatch. The result is not guaranteed reconciliation. It is localization of what has and has not been reconciled.

10. Relationality is not arbitrariness

A common objection is that scope-indexing is “just relativism.” That phrase is not a frame-free verdict; it is itself an adjudicative classification. If “relativism” means that no claim can be more warranted than any other, the present framework rejects it: proof, consistency, counterexample, exact recovery, reproducibility, calibration, and receipt quality can distinguish claims. If “relativism” merely means that describable determinations are expressed through relations, the label renames rather than refutes the proposition. If it is instead asserted as an absolute defeater of relation-indexed claims, it owes a warrant for the exemption by which its own adjudication escapes the relation-dependence it condemns.

A relation may be exact, invariant, reproducible, formally necessary, or empirically constrained. Calling a result relational specifies the structure through which it is warranted; it does not diminish the warrant. Relationality is therefore not offered here as a universal solvent or as a covert ontology in which “only relations are real.” It is a discipline imposed on the paper’s own assertions: the relation doing the adjudicative work must remain available for inspection when the resulting predicate is disputed.

11. The double edge of critique

The framework grants itself no protected standpoint. A criticism is another claim and must survive the inferential discipline it invokes. Let criticism C target proposition P by alleging defect F . If the derivation of C itself exhibits F under the same relevance criteria, C cannot function as a decisive defeater unless an independently warranted distinction licenses the asymmetry.

$F(P) = \text{reject}(P)$, while $F(C)$, requires the same burden unless a registered distinction explains the difference.

No claim is protected by being the target, and no criticism is protected by being the criticism.

This is not immunity from criticism. A critic may show that a definition changes scope, that a bridge fails to preserve the target, that a tail is already closed, that an alleged constitutive dependency is dispensable, that an enlargement introduces irrelevant structure, or that an example equivocates. Any successful criticism changes the record. What the critic may not do without further warrant is use relation-dependence, axiomatic dependence, or semantic compactification as a defeater while exempting the criticism from the same accusation.

The Double-Edged Receipt formalization already supplies the mechanical precedent: accusations of denialism or contrarianism receive no exemption from the receipt rule, and a registered relevant distinction can defeat an apparent asymmetry. [4]

12. REOPEN as target-indexed recovery

REOPEN is not a demand to reconstruct the complete genealogy of every term. It is indexed to the disputed inference I :

$\rho_I(P(x)) \mapsto \text{target-relevant dependencies of } P(x) \text{ for } I.$

The operation stops when enough of the suppressed construction has been recovered to adjudicate the inference at issue. If a deeper dependency is material, reopening continues. If it is not material, no infinite regress is required.

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13. Principal propositions

Proposition 1 — Scope non-amplification

If $\Gamma \vdash P(x)$, semantic compactification to $P(x)$ does not by itself license $P(x)$ at a stronger registry Γ' . A stronger use requires preservation of the target-relevant dependencies or an independently warranted bridge.

Proposition 2 — Tail non-equivalence

Traversal, closure, and enlargement are distinct. Additional traversal does not in general entail universal closure; unresolved traversal does not entail impossibility of closure; enlargement does not entail improvement unless the added distinction is shown relevant to the target.

Proposition 3 — Foundational non-promotion

From $A \vdash T$, successful derivation does not by itself establish the finality, uniqueness, or exhaustiveness of A as a basis for T .

Proposition 4 — Reflexive equal-burden

A criticism that treats defect F as disqualifying cannot exempt its own derivation from F without an independently supported distinction that makes the asymmetry relevant.

Proposition 5 — Recoverable compactification

Semantic compactification is compatible with rigorous cumulative reasoning when the dependencies material to disputed inference remain recoverable. The framework therefore does not oppose objectification as an operation; it opposes silent warrant amplification through irreversible or scope-erasing reuse.

14. Limits

- The framework does not supply a universal ontology or a final theory of representation. It neither asserts nor denies an intrinsic, observer-independent object reality. Absence of a receipt for intrinsicness is not a receipt for nonexistence of intrinsicness; nor does conceivability of intrinsicness establish it.
- It does not claim that every dispute is resolvable by semantic reopening.
- It does not infer that every formal system is incomplete in the technical Gödelian sense.
- It does not infer motive, psychology, bad faith, or identity from structural adjudicative failures.
- It does not make every enlargement legitimate; added distinctions require their own receipts.
- It does not weaken exact mathematical closure. A theorem closes the obligation it actually proves.
- It does not claim that local self-distinction is sufficient for consciousness; that example establishes only that local discrimination does not logically require prior global closure.

15. Conclusion

Complex thought requires blocks. Objects, predicates, classes, theorems, measurements, and named phenomena allow prior work to be carried forward without perpetual reconstruction. The mistake is not compactification. The mistake is allowing the compact form to acquire a stronger warrant merely because the construction that earned it has disappeared from view. The same rule binds this paper: because it makes reasoned assertions from here, it must not flatten its own conditions of assertion into a claim to speak from nowhere.

Tail predicates make the distinction especially visible. Some tails are extensional: unresolved members remain within a declared class. Others are constitutive: recursive inquiry reaches a dependency participating in construction of the very distinction under examination. Either can be traversed, closed, enlarged, compared, or left OPEN, but none is licensed to become the opposite polarity merely because current adjudication stops at its edge.

The same discipline applies at the highest point a proof can presently reach. A theorem may be exact. An axiom system may be extraordinarily powerful. A formal receipt may be decisive. None is

diminished by refusing to promote successful consequence into an unearned declaration of finality. The mountain remains a mountain; its summit need not be treated as a platform outside the continuum from which every possible mountain has already been surveyed.

**Proof certifies consequence, not finality. Relation-indexing does not weaken exactness.
Critique receives no exemption from the rule it invokes.**

The practical rule is therefore not to abolish object-language or demand a universal language. It is to preserve recoverable scope: compactify freely, carry the coordinates that matter, require a bridge when scope changes, and reopen the block when a disputed inference depends upon what the block suppressed.

References and formal antecedents

[1] EmergenceByDesign / MetaLabe. Λ at the Point of Adjudication: Conveyance, carried conditions, persistence, and the Navier-Stokes adjudication instrument. Successor laboratory edition R4, 14 September 2026. Formal packet reports 278 checked theorem declarations and includes the finite-window, all-stage, recovery, π/e , and scoped adjudication results used here.

[2] EmergenceByDesign. Astra 15A — dimensional enlargement. Browser entry B1. Exact rational arithmetic instrument for Euclidean admissibility, exact embedding rank, orientation ambiguity under reflection, and transport of one supplied sign.

https://www.emergencebydesign.com/tools/dimensional_enlargement/Astra_15A_demonstrator.html

[3] EmergenceByDesign / MetaLabe. Deflation of Pharmakon-Based Delusion. Lean receipt packet, 9 September 2026. Finite semantic-deflation model used here for lexical identity versus predicate identity.

[4] EmergenceByDesign / MetaLabe. Denialism and Contrarianism: A Double-Edged Receipt Analysis of Asymmetric Negative Promotion. Technical Note v0.1. Defines structural denialism and contrarianism, negative-promotion symmetry, non-vindication, exchange, restoration, and reflexive liability.

Appendix A. Formalization targets for the companion receipt

The following statements are candidates for a dedicated Lean companion. They should be encoded narrowly enough that successful checking cannot be mistaken for stronger philosophical conclusions.

- Scope non-amplification as a bridge-indexed transport theorem/countermodel.
- Extensional tail obstruction using finite-window opposite extensions, imported or re-proved from Λ .
- General residual-obligation datatype preserving nested quantifier structure rather than forcing every tail into set difference.
- Constitutive-tail witness: removing a registered dependency makes a target formula or discriminator unavailable in the selected encoding; interpret only as dependence of that encoding, not absolute necessity.
- Foundational non-promotion countermodel: A entails T while an alternative basis A' also entails T , defeating uniqueness from consequence alone.

- Reflexive equal-burden theorem parameterized by a defect predicate F and a licensed relevance distinction.
- REOPEN adequacy theorem indexed to a target/inference: recovered coordinates suffice exactly when the target factors through the recovered representation.

Appendix B. MetaLabe semantic workbench preservation notes

The interactive ideas developed alongside this paper are preserved as future MetaLabe Full Semantic Workbench functions rather than requirements for a single companion instrument. A compact statement should be progressively reopenable through OBJECT $\rightarrow$ PREDICATE $\rightarrow$ WITNESS/ADJUDICATOR $\rightarrow$ CONDITIONS $\rightarrow$ SCOPE $\rightarrow$ EVIDENCE $\rightarrow$ OPERATIONS $\rightarrow$ QUANTIFIERS $\rightarrow$ AXIOMS/PRIMITIVES $\rightarrow$ RESIDUAL OBLIGATION, with a reversible path back to the useful compact expression.

Preserved workbench cases include: finite traversal versus theorem-level closure; Astra-style mixed fibre and dimensional enlargement; semantic-deflation label identity versus predicate identity; double-edged negative promotion with OPEN as a distinct adjudicative status; constitutive-tail reopening of enabling commitments; and the simple red-ball control in which object, illumination, sensor, classification, and intervention can be varied independently. The workbench should allow granular stepwise operations and optional formal receipts rather than forcing the entire formal registry onto the user at once.